

RISANA BONNIA MALULEKE

Final Year Computer Systems Engineering Student

CONTACT INFORMATION

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PROFESSIONAL SUMMARY

Computer Systems Engineering student with hands-on experience developing full-stack web applications, e-commerce platforms, database-driven systems, REST APIs and IoT solutions. Proficient in Python, Flask, JavaScript, C++, HTML, CSS, MySQL and SQLite, with practical experience in authentication, role-based access control, Git/GitHub and production deployment. Experienced with Linux servers, Nginx, Gunicorn, Cloudflare, embedded systems and networking. Entrepreneurial experience providing technical support, troubleshooting hardware/software issues and working directly with clients. Seeking opportunities in software development, IT support, DevOps, cloud technologies, networking or graduate/internship programmes.

EDUCATION

Vaal University of Technology | Vanderbijlpark, South Africa

Diploma in Electrical Engineering (Computer Systems)

2023 – Present

RELEVANT AREAS

Programming (C++, Python) • Computer Systems • Networking • Database Systems • Embedded Systems • Digital Systems • Operating Systems • Web & Software Development

TECHNICAL SKILLS

Programming & Web Development: Python, C++, JavaScript, HTML, CSS, Flask, REST APIs

Databases: MySQL, SQLite, Database Design, SQL

Software Development: Git, GitHub, Authentication, Role-Based Access Control, API Integration

Deployment & Infrastructure: Linux, Ubuntu Server, Nginx, Gunicorn, Cloudflare, DNS, SSL

Networking: TCP/IP, Subnetting, Network Configuration, Windows OS Management

Embedded & IoT: Arduino, ESP32, Sensors, Serial Communication, Real-Time Data Systems

Other: Technical Support, Troubleshooting, Client Relations, Problem-Solving, Business Management

PROJECTS

FULL-STACK IoT SMART HOME AUTOMATION & SECURITY SYSTEM

Vaal University of Technology Project | May 2025

- Engineered a full-stack IoT application using Python Flask, MySQL and Bootstrap to monitor and visualize sensor data from embedded systems.
- Implemented user authentication and role-based access control with customized dashboards for administrators and users.
- Developed REST API endpoints for sensor data ingestion and retrieval.
- Implemented real-time visualization using Chart.js with 2-second polling intervals.
- Integrated a Telegram notification bot for motion detection and critical environmental alerts.
- Developed a multi-threaded architecture for concurrent serial communication, database operations and web requests.
- Built responsive interfaces displaying temperature, humidity, motion and light sensor data.
- Worked with Arduino/ESP32, sensors, serial communication, databases and web technologies.

HOTSHOT FABRICS — Full-Stack E-Commerce Platform

Full-Stack Developer | Production Project

- Developed and deployed a production e-commerce platform using Python, Flask, HTML, CSS, JavaScript and MySQL.
- Implemented product management, shopping workflows and database-driven functionality.
- Integrated PayFast payment processing for online transactions.
- Deployed the application on an Ubuntu VPS using Nginx and Gunicorn.
- Configured production domain, DNS and SSL using Cloudflare.
- Managed environment variables, application configuration and production database operations.
- Applied full-stack development practices across frontend, backend, database and server infrastructure.

SASNEWS | South African News & Current Affairs

Full-Stack Web Application | Ongoing

- Developing a database-driven news platform using Flask, MySQL, HTML, CSS and JavaScript.
- Implementing user authentication, staff roles, article management and commenting functionality.
- Developing administrative dashboards for managing news content and website operations.
- Building newsletter functionality and email integration.
- Designing the application architecture for production deployment.
- Working across backend development, database design, frontend development and system administration.

CERTIFICATIONS

- **CCNAv7:Introduction to Networks** | Cisco Networking academy | July 2024
- **NDG Linux Essentials** | Cisco Networking academy | July 2024
- **CCNA: Switching, Routing, and Wireless Essentials** | Cisco Networking Academy | April 2026

ADDITIONAL EXPERIENCE & COMPETENCIES

- Full-Stack Web Development
- Database Development & Management
- Linux Server Administration
- Networking & Technical Support
- Embedded Systems & IoT Development
- Git/GitHub Development
- Client Communication & Technical Problem-Solving
- Business & Project Management

ENTREPRENEURIAL EXPERIENCE

Founder & Technician | RBM Gadgets Solution

August 2024 – Present

- Manage a technology support and repair service.
- Diagnose and resolve hardware and software problems.
- Install and troubleshoot Windows operating systems.
- Perform mobile-device repairs and technical diagnostics.
- Communicate directly with clients to understand technical requirements.
- Manage projects, customer relationships and problem resolution.

REFERENCES

Available upon request